GRAND ROUNDS CASE · FOLLOW-UP PROGRAM

Followup 26-16

Hypercapnic respiratory failure and arrest in severe obesity

Patient	A 45-year-old female weighing 260 kg, with obesity hypoventilation syndrome, sleep apnea treated with positive airway pressure at home, heart failure with an ejection fraction of 40 percent, persistent atrial fibrillation on apixaban, hypothyroidism, mild asthma, and prior gastric surgery. **She had been discharged eleven days earlier after about two months in hospital, including two weeks intubated and ventilated prone.**
Outcome	She stopped breathing during loading, with a weak pulse, and lost her pulse about two minutes before arrival. Circulation returned in the emergency department. **The hospital diagnosed acute hypoxic and hypercapnic respiratory failure from multifocal pneumonia and large pleural effusions, with shock.** On hospital day 1 her endotracheal tube failed when she was turned from prone to supine, and she arrested again minutes after it was replaced.
Disposition	Died in intensive care on hospital day 1. After 20 minutes of CPR, her family asked for comfort measures. The medical examiner was notified and did not perform an autopsy.

The call

The call was dispatched as breathing problems. The crew found her in bed on her left side, alert, short of breath, with mild wheezes throughout. She told the crew she had recently been discharged and was waiting for home oxygen that had not arrived. This was her third call; on the first two she had a breathing treatment and declined transport.

Oxygen and albuterol brought her saturation to 100 percent, and she felt better. The crew told her this was her third call and she was not improving, and she agreed to go. Fire responded to help move her, and she was placed on the cot on her side, in the position she lay in. While she was being loaded into the ambulance she said she felt hot, then became unresponsive, with a weak pulse and possible respiratory arrest.

The crew ventilated her and placed an airway and a humeral IO, with monitor leads on her back. She lost her pulse shortly before arrival. Epinephrine was given on arrival, and compressions began on the emergency department cart; the narrative records that they could not be performed on the cot because of her size and position.

Times below are from patient contact.

From contact	Finding or action
+1 min	Pulse 108, respirations 22, SpO2 77% on room air; blood pressure unobtainable
+2 min	Oxygen 4 LPM by nasal cannula
+4 min	Albuterol 2.5 mg by nebulizer; SpO2 rises to 100%
+30 min	Unresponsive during loading, weak pulse; bag-valve-mask ventilation at 15 LPM

From contact	Finding or action
+32 to +36 min	Nasopharyngeal airway, i-gel, cardiac monitor, right humeral IO
+38 min	Leaves scene
+39 min	Pulse lost, about two minutes before arrival
+41 min	Arrives at the emergency department
+42 min	Epinephrine 1 mg IO

The numbers to hold onto are a saturation of 100 percent about 25 minutes before she stopped breathing, and a carbon dioxide of 90 on her first blood gas.

Primary impression

The crew's impression was respiratory arrest. The hospital agreed and named the cause: acute hypoxic and hypercapnic respiratory failure from multifocal pneumonia and large pleural effusions, on top of obesity hypoventilation syndrome. She had declined transport twice before. On the third call this crew recognized that a patient calling repeatedly for the same complaint was not recovering, said so to her, and got her to agree to go. That was the right reading of the call.

Hospital course

Emergency department, day 0. She arrived pulseless with the i-gel in place. The humeral IO was lost during the move to the emergency department cart, and a tibial IO was placed. Circulation returned after the first round of CPR and epinephrine, lasted about four minutes, and returned again after two more rounds. She was intubated on the first attempt by video laryngoscopy, through a grade IV view. Her first arterial gas showed a pH of 7.08, a PCO2 of 90, and a PO2 of 44. CT angiography showed no pulmonary embolism, multifocal pneumonia worse on the right, moderate bilateral pleural effusions, suspected mucus plugging in both lower lobes, and fractures of right ribs 4 to 6 and left ribs 2 to 7 of indeterminate age. She was placed semi-prone to improve ventilation, started on norepinephrine and antibiotics, and admitted to intensive care. Her next gas showed a PCO2 of 54 and a pH of 7.31.

Hospital day 1. She was ventilated prone on a small dose of norepinephrine. She was turned supine for wound checks and an echocardiogram, a cuff leak appeared immediately, and her saturation fell. The tube was exchanged over a bougie on the first attempt and confirmed with end-tidal CO2 and bronchoscopy, and her saturation recovered into the 90s. A few minutes later she went into atrial fibrillation with a rapid ventricular response, her blood pressure became unobtainable, and she lost her pulse. She received 20 minutes of CPR in PEA. Her brother, at the bedside and in contact with the family, asked for comfort measures, and she died. The medical examiner did not perform an autopsy.

Final diagnoses: Cardiac arrest. Acute hypoxic and hypercapnic respiratory failure from multifocal pneumonia and large pleural effusions. Shock. Lactic and metabolic acidosis.

Education points

Obesity hypoventilation syndrome

Let's start with what the condition is. Obesity hypoventilation syndrome is obesity with a carbon dioxide above 45 mmHg while awake, with no other cause such as COPD or neuromuscular disease.

The weight of the chest wall and abdomen stiffens the respiratory system and shrinks each breath, so the work of breathing rises. Over years, the brainstem's response to carbon dioxide blunts, nights of obstructed breathing leave CO2 higher each morning, and the kidneys hold bicarbonate to buffer it. The patient lives with a carbon dioxide that would make most people drowsy, and with almost no ventilatory reserve.

Anything that adds load tips that balance. Her CT showed multifocal pneumonia, effusions compressing both lungs, and mucus plugging, on top of worsening heart failure. When ventilation falls behind production, carbon dioxide rises. Rising carbon dioxide sedates, sedation reduces breathing further, and the spiral ends in respiratory arrest.

Oxygen saturation and carbon dioxide

A pulse oximeter measures how much hemoglobin carries oxygen. It measures nothing about carbon dioxide, and in a patient breathing supplemental oxygen it can stay normal while ventilation fails.

The alveolar gas equation shows why. The oxygen available in the alveoli is roughly the inspired oxygen minus the carbon dioxide divided by 0.8. On room air, a PCO2 of 90 leaves an alveolar oxygen near 40 mmHg, which is severe hypoxemia, and a saturation in the 70s. Four liters by nasal cannula raises the inspired oxygen to roughly 36 percent, and the same PCO2 of 90 now leaves an alveolar oxygen near 145, enough for a saturation close to 100 percent.

In obesity hypoventilation, supplemental oxygen can also raise carbon dioxide directly. In one randomized study of 24 patients, breathing 100 percent oxygen for 20 minutes raised carbon dioxide by an average of 5 mmHg, and the test was stopped early in three patients because it rose by 10 or more within 16 minutes. In another, oxygen at 28 and 50 percent kept saturations at 98 to 100 percent while carbon dioxide rose and the blood became acidic. Oxygen sends more blood to poorly ventilated lung, where it cannot unload carbon dioxide, and slows breathing.

Her first gas was drawn after a cardiac arrest, which raises carbon dioxide on its own, so it does not show her level on scene. For any patient with known chronic hypercapnia, the practical response is a saturation target of 88 to 92 percent, and end-tidal CO2 by nasal cannula as the ventilation monitor while the patient is still breathing.

Position as physiology

Lying flat, the abdomen of a patient this size pushes the diaphragm up into the chest. The air left in the lungs at the end of a breath falls below the volume at which small airways close, those airways close, and blood passes through lung that is not ventilated. The upper airway also collapses more easily on the back. Some patients with severe obesity cannot tolerate lying flat at all, and lying on the side or prone unloads the diaphragm.

She lay on her side or prone by habit, she was transported on her side, and the hospital ventilated her prone and semi-prone. On hospital day 1 the move from prone to supine produced a cuff leak and a fall in saturation within seconds, and she arrested minutes later.

Moves also shift devices: flexing or extending the neck moves an endotracheal tube several centimeters, and a turn can pull a cuff up to the cords.

Further reading on obesity hypoventilation syndrome. How severe obesity produces chronic hypercapnia, and why oxygen alone does not treat it. https://pmc.ncbi.nlm.nih.gov/articles/PMC2645254/

Further reading on oxygen-induced hypercapnia. The four ways supplemental oxygen raises carbon dioxide, including loss of hypoxic vasoconstriction and the Haldane effect. https://www.monaldi-archives.org/index.php/macd/article/view/2399

Prehospital considerations

A patient who calls repeatedly for the same complaint is not recovering. This crew used that evidence to get a reluctant patient to the hospital.

The interventions

PREHOSPITAL INTERVENTIONS

Oxygen 4 LPM by nasal cannula and albuterol 2.5 mg by nebulizer. Albuterol relaxes bronchial smooth muscle through beta-2 receptors, and it improved her wheezing.

Bag-valve-mask ventilation at 15 LPM. Started when she became unresponsive with a weak pulse.

Nasopharyngeal airway and i-gel. A 260 kg patient on her side is one of the hardest airway problems there is. The crew escalated from an adjunct to a supraglottic airway, and the emergency department found it in place on arrival.

Right humeral IO and epinephrine 1 mg. Access in a patient this size is difficult, and the humeral site worked. Epinephrine went through it on arrival, after she lost her pulse.

IN-HOSPITAL INTERVENTIONS

Endotracheal intubation. First attempt by video laryngoscopy with a hyperangulated blade through a grade IV view, meaning no glottic structures were visible by direct sight.

Tube exchange over a bougie. After the cuff leak on day 1, on the first attempt, with placement confirmed by end-tidal CO2 and bronchoscopy.

The labs, and what they mean

Reference ranges vary between institutions and analyzers. These are the ranges printed on the source reports.

Blood gases and lactate

Lab	First	Later	Reference range
pH, arterial	7.08	7.31	7.35 to 7.45
PCO2, arterial (mmHg)	90	54	35 to 48
PO2, arterial (mmHg)	44	106	83 to 108
Base excess (mmol/L)	-4.9	0.3	-2.0 to 3.0
Lactic acid (mmol/L)	9.8	7.6	0.0 to 2.0

A PCO2 of 90 drove a pH of 7.08. Ventilated through a tube, her PCO2 fell to 54 and her pH rose to 7.31. The base excess is the part to read carefully. A lactate of 9.8 should produce a large base deficit, and hers was only -4.9. Her kidneys had been holding bicarbonate for years to buffer chronic hypercapnia, and her loop diuretic adds bicarbonate too. Acute respiratory acidosis, lactic acidosis, and an underlying metabolic alkalosis were running at once.

Heart failure and clotting markers

Lab	Value	Reference range
Pro-BNP (ng/L)	5,474	125 or less
hs troponin T, baseline (ng/L)	16	Less than 15
hs troponin T, 2 hour (ng/L)	109	Less than 15
D-dimer (ng/mL FEU)	31,640	0 to 500

Her pro-BNP was more than twice her baseline of about 2,000. The troponin rise followed hypoxia, arrest and compressions, and the D-dimer rises with pneumonia and recent hospitalization; CT showed no embolism.

A loop diuretic can hide a lactic acidosis. Her chloride was 89 (98 to 107) and her sodium 132 (136 to 145). Loop diuretics remove chloride with water, and the kidney holds bicarbonate in its place, which is the alkalosis described above.

Small, pale red cells after gastric surgery point to iron deficiency. Her hemoglobin was 9.0 (12.0 to 16.0), with a mean cell hemoglobin concentration of 27.2 (32.0 to 37.0) and a red cell distribution width of 18.8 (11.0 to 16.0). Iron deficiency is on her problem list, and the anemia lowered the oxygen her blood could carry.

Learning points

1. **A saturation of 100 percent on oxygen says nothing about carbon dioxide.** At an inspired oxygen near 36 percent, a PCO2 of 90 still allows a saturation close to 100.
2. **In obesity hypoventilation, supplemental oxygen can raise carbon dioxide.** The saturation target for any patient with chronic hypercapnia is 88 to 92 percent.
3. **Capnography is the ventilation monitor for a patient who is still breathing.** A rising end-tidal CO2, a slowing rate, or drowsiness is hypoventilation.
4. **In severe obesity, position is physiology.** Lying flat collapses lung volume and the airway. A turn to supine produced a tube failure and an arrest within minutes.
5. **Every move threatens every device, and in severe obesity every move is a physiologic event.** Keep the patient in the position she tolerates, move once, have ventilation ready, and recheck airway and access afterward.

Prepared from records obtained through the follow-up request process and de-identified for education. To request follow-up on a call you ran, fill out the request form at https://www.berrienmca.org/followup/